KCE PATHWAYS · GRADE 3 · UNIT 6: BUILDING TOGETHER · WEEK 28 · MATH

The Pie Bakery in Lancaster County — Teacher Lesson Plan & Answer Key

Learning objectives: Students write whole numbers as fractions (4/4 = 1, 12/4 = 3, 3 = 3/1), locate fractions greater than 1 on a number line, compare fractions greater than 1 with the same denominator or the same numerator and with whole numbers, split a fraction greater than 1 into whole pies and leftover slices, and solve two-step fraction problems.

Duration: 35–45 minutes
Materials: Computer or tablet with the lesson open · pencil · printed Pre & Post sheet

Lesson flow (as it appears on screen)

1. Title and name: the student types first and last name so results go to KingCretot Experience. Read the Pathway Points rules together: 2 tries per question, points only for a correct first try, and the answer and reason appear after a second miss. Introduce Ms. Della's pie bakery in Lancaster County, Pennsylvania, with her helpers Deshawn, Danica, Declan and Davina.
2. Warm-Up: Week 27 Review (10 pts): area and perimeter design, same perimeter or same area, and combined rectangles. If a student misses both, review "perimeter goes around; area covers the inside."
3. Picture It: The Pie Bakery in Lancaster County. Pies are cut into 4, 6 or 8 equal slices, and by the end of the day the helpers count more slices than one pie holds. Lancaster Central Market still sells food on land set aside in 1730. Ask: what happens when you have more slices than one pie holds?
4. Word Wrangler: 6 vocab cards (numerator, denominator, whole number, fraction greater than 1, equivalent, number line).
5. Pencil Break (Pre): the student opens or prints the Pre/Post sheet (grade3/prepost/KCE_Math_PrePost_G3W28.pdf) and completes Page 1 (Pre) before any teaching. Do not help with answers yet; it shows what they already know.
6. Whole Pies Made of Slices (10 pts): count by the denominator to find whole pies (24/8 = 3) and write whole numbers as fractions (5 = 15/3). Watch for the mix-up that 5/1 means one fifth; it means 5 wholes.
7. Past 1 on the Number Line (10 pts): count jumps from 0 to name a point (7/4 is 3 jumps past 1) and find which two whole numbers a fraction sits between. Some items use the number line on screen.
8. Comparing Fractions Greater Than 1 (10 pts): same denominator, compare the numerators; same numerator, the bigger pieces win (5/3 > 5/4); compare with a whole number by writing it as a fraction (2 = 8/4). Answers are typed as <, > or =.
9. Whole Pies and Leftover Slices (10 pts): 11/4 is 2 whole pies and 3/4 more, and going the other way, 2 wholes and 1/3 more is 7/3.
10. Bakery Challenge: Two-Step Fraction Problems (10 pts): count only the whole pies, find how many more slices are needed, and add or compare fractions with the same denominator. Ask first: how many slices make one whole pie?
11. Bonus Practice: The Cooling Rack (XP only, not scored).
12. Match-Up Round (10 pts, 2 per first-try match): Set 1 matches fraction vocabulary to meanings, Set 2 matches fractions to the whole numbers they equal, Set 3 matches fractions greater than 1 to wholes and leftover parts.

13. Fix the Sentence (10 pts, 5 each): typed answers are checked exactly (capital letters for names of people and places and the title Ms., end mark, list commas, and one verb fix); curly quotes and extra spaces are forgiven. Both serial-comma and no-serial-comma list forms are accepted. Remind the student to keep the same words in the same order.
14. Speed Arena and Keyboard Corral (XP only, not scored).
15. Boss Battle, Part 1: The Saturday Rush (10 pts, 5 each): counting slices across pies, comparing 8/3 and 8/4, which fraction is between 2 and 3, and a find-the-mistake item on the number line (8/3 is between 2 and 3).
16. Boss Battle, Part 2: Closing Time (10 pts, 5 each): comparing with 2, putting fractions in order, and find-the-mistake items where the student types the correct number (6/2 = 3, not 12; 16/4 is 4 whole pies).
17. Pencil Break (Post): the student turns to Page 2 (Post) of the same sheet (grade3/prepost/KCE_Math_PrePost_G3W28.pdf) and completes it, plus the ★ Challenge if time allows, before seeing the score. The sheet buttons let the student open or print it again.
18. Results (Pie Day Complete!): the score out of 100 is sent to KingCretot Experience. 85 or higher shows “Bakery Hero!” with the congratulations message “Bakery hero! You counted every slice and every pie!” and a big confetti burst; below 85 shows “Great Effort Today!” with encouragement (“Every baker gets better with practice.”), and the next attempt brings a fresh question set (sets rotate 1, 2, 3). Both messages remind the student to finish the Post-Lesson Challenge sheet.

Lesson answer key with reasoning

Every scored item the student sees, in order, across all three rotating question sets. Each item allows 2 tries and earns Pathway Points only on a correct first try. The "why" is the same reasoning shown on screen after a second missed try.

WARM-UP: WEEK 27 REVIEW (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
w1_s0	Set 1: Two steps! The bakery’s rectangular cooling-rack corner has a perimeter of 20 feet and is 4 feet wide. What is its area, in square feet?	**24** (also accepted: 24 square feet, 24 sq ft, 24 square foot)	Step 1: 20 − 4 − 4 = 12 feet for the two lengths, so each length is 6 feet. Step 2: 4 × 6 = 24 square feet.
w2_s0	Set 1: Each rectangle below covers 28 square feet. Which one needs the least border around it? [4 feet by 7 feet / 2 feet by 14 feet / 1 foot by 28 feet / They all need the same border.]	**4 feet by 7 feet**	The perimeters are 22 feet for 4 by 7, 32 feet for 2 by 14, and 58 feet for 1 by 28. The rectangle closest to a square needs the least.
w1_s1	Set 2: The bakery counter is split into a 6 by 10 part and a 6 by 3 part. What is the total area of the counter, in square feet?	**78** (also accepted: 78 square feet, 78 sq ft, 78 square foot)	6 × 10 = 60 and 6 × 3 = 18. 60 + 18 = 78 square feet. That is the same as 6 × 13 = 78.
w2_s1	Set 2: Find the mistake. Davina says two rectangles with the same perimeter always have the same area. What is wrong? [A 2 by 6 and a 4 by 4 rectangle both have a perimeter of 16 feet, but their areas are 12 and 16 square feet. / Nothing is wrong. Same perimeter always means same area. / Same perimeter always means the area is half the perimeter. / Rectangles cannot have the same perimeter.]	**A 2 by 6 and a 4 by 4 rectangle both have a perimeter of 16 feet, but their areas are 12 and 16 square feet.**	Same perimeter does not mean same area: 2 + 6 + 2 + 6 = 16 and 4 + 4 + 4 + 4 = 16, but 2 × 6 = 12 and 4 × 4 = 16.
w1_s2	Set 3: The kitchen floor is a 9 by 7 rectangle with a 3 by 2 corner cut out for a closet. What is the area of the floor that is left, in square feet?	**57** (also accepted: 57 square feet, 57 sq ft, 57 square foot)	Whole rectangle: 9 × 7 = 63. Corner: 3 × 2 = 6. 63 − 6 = 57 square feet.

ID	Question	Answer	Why (explain to student)
w2_s2	Set 3: A 5 by 7 rectangle has a perimeter of 24 feet. Which rectangle has the same perimeter? [3 by 9 / 4 by 9 / 5 by 5 / 2 by 7]	**3 by 9**	3 + 9 + 3 + 9 = 24 feet. The others are 26, 20, and 18 feet.

WHOLE PIES MADE OF SLICES (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
a1_s0	Set 1: Each cherry pie is cut into 8 equal slices. The shelf holds 24/8 of a pie. How many whole pies is 24/8?	**3** (also accepted: 3 wholes, 3 whole pies, 3 pies, 3 whole, three)	Count by 8s: 8, 16, 24. That is 3 counts, so 24/8 = 3 whole pies.
a2_s0	Set 1: Which fraction is NOT equal to 1? [6/1 / 6/6 / 3/3 / 1/1]	**6/1**	6/1 means 6 wholes that are each 1 piece, so 6/1 = 6. The others have the same numerator and denominator, so each equals 1.
a1_s1	Set 2: Two steps! Ms. Della bakes 5 pumpkin pies and cuts each one into thirds. How many thirds does she have in all?	**15** (also accepted: 15 thirds, 15/3, 15 / 3, fifteen)	Step 1: Each pie has 3 thirds. Step 2: 5 × 3 = 15 thirds, so 5 = 15/3.
a2_s1	Set 2: Find the mistake. Declan says 5/1 means one fifth of a pie. What is wrong? [5/1 means 5 wholes that are each 1 piece, so 5/1 = 5 pies. One fifth is 1/5. / Nothing is wrong. 5/1 is one fifth. / 5/1 means 5 halves of a pie. / 5/1 is the same as 0, because 5 − 5 = 0.]	**5/1 means 5 wholes that are each 1 piece, so 5/1 = 5 pies. One fifth is 1/5.**	The denominator 1 means each whole is in 1 piece, and the numerator 5 counts 5 of those wholes. So 5/1 = 5.
a1_s2	Set 3: Type the missing numerator: 7 =?/2. How many halves make 7 whole pies?	**14** (also accepted: 14 halves, 14/2, 14 / 2, fourteen)	Each pie has 2 halves. 7 × 2 = 14, so 7 = 14/2.
a2_s2	Set 3: The bakery has 20 fourths of shoofly pie on the shelf. How many whole pies is that?	**5** (also accepted: 5 wholes, 5 whole pies, 5 pies, 5 whole, five)	Count by 4s: 4, 8, 12, 16, 20. That is 5 counts, so 20/4 = 5 whole pies.

PAST 1 ON THE NUMBER LINE (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
b1_s0	Set 1: This number line goes from 0 to 3 in fourths. What fraction names point A?	**7/4** (also accepted: 7 / 4, 7 fourths, seven fourths)	Count the jumps of 1/4 from 0. Point A is 7 jumps from 0, so it is 7/4.
b2_s0	Set 1: Between which two whole numbers is 11/4? [between 2 and 3 / between 1 and 2 / between 0 and 1 / between 3 and 4]	**between 2 and 3**	2 = 8/4 and 3 = 12/4. 11 is between 8 and 12, so 11/4 is between 2 and 3.
b1_s1	Set 2: This number line goes from 0 to 3 in thirds. What fraction names point B?	**5/3** (also accepted: 5 / 3, 5 thirds, five thirds)	Each whole has 3 jumps. Point B is 5 jumps from 0, so it is 5/3: 2 jumps past 1.
b2_s1	Set 2: Two steps! Declan's dough strip reaches 5/4 feet. How many more fourths of a foot must he roll to reach 2 feet?	**3** (also accepted: 3 fourths, 3/4, 3 / 4, three)	Step 1: 2 feet = 8/4 feet. Step 2: 8 − 5 = 3, so he needs 3 more fourths of a foot.
b1_s2	Set 3: This number line goes from 0 to 3 in fourths. Type the fraction with a denominator of 4 that names point C.	**10/4** (also accepted: 10 / 4, 10 fourths, ten fourths)	2 = 8/4, and point C is 2 jumps past 2. That is 10/4.

ID	Question	Answer	Why (explain to student)
b2_s2	Set 3: Which explanation is correct? Why is 9/4 greater than 2? [2 is the same as 8/4, and 9 fourths is 1 more fourth than 8 fourths. / Because 9 is greater than 4. / Because 9/4 has a bigger denominator than 2. / It is not. 9/4 is less than 2.]	**2 is the same as 8/4, and 9 fourths is 1 more fourth than 8 fourths.**	9 > 4 only shows that 9/4 is more than 1. To compare with 2, write 2 as 8/4. Since 9 > 8, 9/4 > 2.

COMPARING FRACTIONS GREATER THAN 1 (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
k1_s0	Set 1: Compare. Type <, >, or =. 7/3? 5/3	**>** (also accepted: greater than, is greater than, greater)	The denominators are the same, so compare the numerators: 7 thirds is more than 5 thirds. 7/3 > 5/3.
k2_s0	Set 1: Compare. Type <, >, or =. 6/3? 2	**=** (also accepted: equal, equals, equal to, is equal to)	Count by 3s: 3, 6. That is 2 wholes, so 6/3 = 2.
k1_s1	Set 2: Compare. Type <, >, or =. 5/4? 5/6	**>** (also accepted: greater than, is greater than, greater)	The numerators are the same. Fourths are bigger pieces than sixths, so 5 fourths is more. 5/4 > 5/6.
k2_s1	Set 2: Find the mistake. Danica says 7/8 > 7/6 because 8 is bigger than 6. What is wrong? [Eighths are smaller pieces than sixths, so 7 eighths is less than 7 sixths: 7/8 < 7/6. / Nothing is wrong. 7/8 > 7/6. / They are equal, because both numerators are 7. / 7/8 is greater, because it is closer to 1.]	**Eighths are smaller pieces than sixths, so 7 eighths is less than 7 sixths: 7/8 < 7/6.**	When the numerators match, the bigger denominator means smaller pieces. 7/8 is less than 1, and 7/6 is more than 1.
k1_s2	Set 3: Two steps! Compare. Type <, >, or =. 9/4? 5/3	**>** (also accepted: greater than, is greater than, greater)	9/4 is more than 2, because 2 = 8/4. 5/3 is less than 2, because 2 = 6/3. So 9/4 > 5/3.
k2_s2	Set 3: All four pies are the same size. Which amount is the most pie? [10/3 / 10/4 / 10/6 / 10/8]	**10/3**	All the numerators are 10. Thirds are the biggest pieces, so 10 thirds is the most pie.

WHOLE PIES AND LEFTOVER SLICES (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
d1_s0	Set 1: Ms. Della has 11/4 pies of shoofly pie, cut in fourths. How many WHOLE pies can she put in boxes?	**2** (also accepted: 2 wholes, 2 whole pies, 2 pies, 2 whole, two)	Count by 4s: 4, 8. That is 2 whole pies, with 3 fourths left over.
d2_s0	Set 1: Deshawn has 2 whole pies and 1/3 of a pie more. How many thirds is that in all? Write it as a fraction.	**7/3** (also accepted: 7 / 3, 7 thirds, seven thirds)	Each whole pie is 3/3. 3 + 3 + 1 = 7 thirds, so it is 7/3.
d1_s1	Set 2: Two steps! A pie is cut into 6 equal slices. Customers buy 15 slices in all. Write how much pie they bought as a fraction with a denominator of 6.	**15/6** (also accepted: 15 / 6, 15 sixths, fifteen sixths)	Step 1: Each slice is 1/6 of a pie. Step 2: 15 slices is 15 sixths, or 15/6. That is 2 whole pies and 3/6 more.
d2_s1	Set 2: Which shows 5/2 of a pie? [2 whole pies and 1/2 of another pie / 5 whole pies and 1/2 of another pie / 1 whole pie and 1/2 of another pie / 1/2 of 1 pie]	**2 whole pies and 1/2 of another pie**	Two halves make one whole. 5 halves = 2 + 2 + 1, so it is 2 whole pies and 1 half more.

ID	Question	Answer	Why (explain to student)
d1_s2	Set 3: Type the missing number: 13/4 =? whole pies and 1/4 more.	**3** (also accepted: 3 wholes, 3 whole pies, 3 pies, 3 whole, three)	Count by 4s: 4, 8, 12. That is 3 whole pies, and 13 − 12 = 1 fourth is left over.
d2_s2	Set 3: Which explanation is correct? Declan says 12/4 and 3 are the same amount. [He is right. 4 fourths make 1 whole, so 12 fourths make 3 wholes. / He is wrong. 12/4 is a fraction, so it must be less than 1. / He is wrong. 12/4 means 12 whole pies. / He is wrong. 12/4 = 4, because 12 − 8 = 4.]	**He is right. 4 fourths make 1 whole, so 12 fourths make 3 wholes.**	Count by 4s: 4, 8, 12. That is 3 groups of 4 fourths, so 12/4 = 3.

BAKERY CHALLENGE: TWO-STEP FRACTION PROBLEMS (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
c1_s0	Set 1: Two steps! Each pie is cut into 8 slices. On Saturday the bakery sells 27 slices. How many whole pies' worth of slices is that? Count only the whole pies.	**3** (also accepted: 3 wholes, 3 whole pies, 3 pies, 3 whole, three)	Step 1: Count by 8s: 8, 16, 24. Step 2: 24 slices make 3 whole pies, and the other 3 slices are only part of a pie.
c2_s0	Set 1: Two steps! Ms. Della needs 3 pies cut into sixths for a party. She has 14 sixths ready. How many more sixths does she need?	**4** (also accepted: 4 sixths, 4/6, 4 / 6, four)	Step 1: 3 pies = 3 × 6 = 18 sixths. Step 2: 18 − 14 = 4 more sixths.
c1_s1	Set 2: Two steps! Davina bakes 4 pies and cuts each one into thirds. She sells 7/3 of a pie. How many thirds are left?	**5** (also accepted: 5 thirds, 5/3, 5 / 3, five)	Step 1: 4 pies = 4 × 3 = 12 thirds. Step 2: 12 − 7 = 5 thirds are left, or 5/3 of a pie.
c2_s1	Set 2: Deshawn has 3 whole pies and 3/4 of a pie more. Write the total as a fraction with a denominator of 4.	**15/4** (also accepted: 15 / 4, 15 fourths, fifteen fourths)	3 whole pies = 12/4. 12 + 3 = 15 fourths, so the total is 15/4.
c1_s2	Set 3: Two steps! Danica has 9/4 pies of apple and 7/4 pies of cherry. How many whole pies is that in all?	**4** (also accepted: 4 wholes, 4 whole pies, 4 pies, 4 whole, four)	Step 1: 9 fourths + 7 fourths = 16 fourths. Step 2: Count by 4s: 4, 8, 12, 16. That is 4 whole pies.
c2_s2	Set 3: Is 11/4 of a pie more or less than 3 whole pies? Compare. Type <, >, or =. 11/4? 3	**<** (also accepted: less than, is less than, less)	3 = 12/4. 11 fourths is 1 fourth less than 12 fourths, so 11/4 < 3.

MATCH-UP ROUND (2 POINTS PER FIRST-TRY MATCH)

ID	Question	Answer	Why (explain to student)
match_s0_1	Set 1: Match numerator	**the top number of a fraction**	2 points if matched on the first try.
match_s0_2	Set 1: Match denominator	**how many equal parts make 1 whole**	2 points if matched on the first try.
match_s0_3	Set 1: Match whole number	**0, 1, 2, 3, and so on**	2 points if matched on the first try.
match_s0_4	Set 1: Match fraction greater than 1	**numerator bigger than denominator**	2 points if matched on the first try.
match_s0_5	Set 1: Match equivalent	**names the same amount**	2 points if matched on the first try.
match_s1_1	Set 2: Match 8/4	**2**	2 points if matched on the first try.
match_s1_2	Set 2: Match 9/3	**3**	2 points if matched on the first try.
match_s1_3	Set 2: Match 8/8	**1**	2 points if matched on the first try.

ID	Question	Answer	Why (explain to student)
match_s1_4	Set 2: Match 12/2	**6**	2 points if matched on the first try.
match_s1_5	Set 2: Match 4/1	**4**	2 points if matched on the first try.
match_s2_1	Set 3: Match 7/4	**1 whole and 3/4 more**	2 points if matched on the first try.
match_s2_2	Set 3: Match 5/2	**2 wholes and 1/2 more**	2 points if matched on the first try.
match_s2_3	Set 3: Match 10/3	**3 wholes and 1/3 more**	2 points if matched on the first try.
match_s2_4	Set 3: Match 9/8	**1 whole and 1/8 more**	2 points if matched on the first try.
match_s2_5	Set 3: Match 13/6	**2 wholes and 1/6 more**	2 points if matched on the first try.

FIX THE SENTENCE (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
fs1_s0	Set 1: Fix the sentence: lancaster county is in pennsylvania	**Lancaster County is in Pennsylvania.**	Lancaster County and Pennsylvania are names of places, so they get capital letters. End the telling sentence with a period.
fs2_s0	Set 1: Fix the sentence: the bakery sells apple pies cherry pies and shoofly pies	**The bakery sells apple pies, cherry pies, and shoofly pies.** (also accepted: The bakery sells apple pies, cherry pies and shoofly pies.)	Start with a capital T, put commas between the things in the list, and end with a period.
fs1_s1	Set 2: Fix the sentence: how many slices are in one pie	**How many slices are in one pie?**	Start with a capital H. This sentence asks something, so it ends with a question mark.
fs2_s1	Set 2: Fix the sentence: deshawn danica and declan rolled the dough	**Deshawn, Danica, and Declan rolled the dough.** (also accepted: Deshawn, Danica and Declan rolled the dough.)	Each name gets a capital letter, commas go between the names in the list, and the sentence ends with a period.
fs1_s2	Set 3: Fix the sentence: did ms. della bake a pumpkin pie	**Did Ms. Della bake a pumpkin pie?**	Start with a capital D. Ms. Della is a name, so it gets capital letters. The sentence asks something, so it ends with a question mark.
fs2_s2	Set 3: Fix the sentence: the pies is cooling on the rack	**The pies are cooling on the rack.**	Start with a capital T and end with a period. Pies means more than one, so use are, not is.

BOSS BATTLE: THE SATURDAY RUSH (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
bb1_s0	Set 1: Two steps! Ms. Della cuts 4 pies into sixths and 1 more pie into thirds. How many slices does she have in all?	**27** (also accepted: 27 slices)	Step 1: 4 pies in sixths: 4 × 6 = 24 slices, so 4 = 24/6. Step 2: 1 pie in thirds is 3 slices, so 1 = 3/3. 24 + 3 = 27 slices.
bb2_s0	Set 1: Which fraction is between 2 and 3 on a number line? [9/4 / 7/4 / 12/4 / 6/4]	**9/4**	2 = 8/4 and 3 = 12/4. Only 9/4 is between them. 12/4 is exactly 3, and 7/4 and 6/4 are between 1 and 2.

ID	Question	Answer	Why (explain to student)
bb1_s1	Set 2: Compare. Type <, >, or =. 8/3? 8/4	**>** (also accepted: greater than, is greater than, greater)	The numerators are the same. Thirds are bigger pieces than fourths, so 8/3 > 8/4.
bb2_s1	Set 2: Two steps! Danica packs 3 boxes. Each box holds 6/3 pies, cut in thirds. How many whole pies does she pack in all?	**6** (also accepted: 6 wholes, 6 whole pies, 6 pies, 6 whole, six)	Step 1: Count by 3s: 3, 6. So 6/3 = 2 whole pies in each box. Step 2: 3 boxes × 2 pies = 6 whole pies.
bb1_s2	Set 3: Type the whole number: 18/3 =?	**6** (also accepted: 6 wholes, 6 whole pies, 6 pies, 6 whole, six)	Count by 3s: 3, 6, 9, 12, 15, 18. That is 6 counts, so 18/3 = 6.
bb2_s2	Set 3: Find the mistake. On a number line from 0 to 3 in thirds, Davina counts 8 jumps from 0 and labels the point 8/3. She says 8/3 is between 1 and 2. What is wrong? [2 = 6/3 and 3 = 9/3, so 8/3 is between 2 and 3. / Nothing is wrong. 8/3 is between 1 and 2. / 8/3 is between 0 and 1, because it is a fraction. / 8/3 sits right on top of 8.]	**2 = 6/3 and 3 = 9/3, so 8/3 is between 2 and 3.**	Count by 3s to find the wholes: 3 thirds is 1, 6 thirds is 2, and 9 thirds is 3. 8 is between 6 and 9, so 8/3 is between 2 and 3.

BOSS BATTLE: CLOSING TIME (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
bb3_s0	Set 1: Find the mistake. Deshawn says 6/2 = 12 because 6 × 2 = 12. Type the correct whole number.	**3** (also accepted: 3 wholes, 3 whole pies, 3 pies, 3 whole, three)	Halves come 2 to a whole. Count by 2s: 2, 4, 6. That is 3 wholes, so 6/2 = 3.
bb4_s0	Set 1: Compare. Type <, >, or =. 11/6? 2	**<** (also accepted: less than, is less than, less)	2 = 12/6. 11 sixths is 1 sixth less than 12 sixths, so 11/6 < 2.
bb3_s1	Set 2: Find the mistake. Danica says 16/4 of a pie is 16 whole pies. Type the correct number of whole pies.	**4** (also accepted: 4 wholes, 4 whole pies, 4 pies, 4 whole, four)	16 counts fourths, not whole pies. Count by 4s: 4, 8, 12, 16. That is 4 whole pies.
bb4_s1	Set 2: Two steps! Ms. Della cuts 5 pies into fourths and sells 13 fourths. Write the pie that is left as a fraction with a denominator of 4.	**7/4** (also accepted: 7 / 4, 7 fourths, seven fourths)	Step 1: 5 pies = 5 × 4 = 20 fourths. Step 2: 20 − 13 = 7 fourths, so 7/4 of a pie is left.
bb3_s2	Set 3: Put these in order from least to greatest: 7/3, 2, 5/3. Type the number that goes in the middle.	**2** (also accepted: 6/3, two)	2 = 6/3. In thirds the list is 5, 6, 7, so the order is 5/3, 2, 7/3. The middle one is 2.
bb4_s2	Set 3: Find the mistake. Declan says 8/8 of a pie is more than 4/4 of a pie because 8 is bigger than 4. Both fractions equal the same whole number. Type that whole number.	**1** (also accepted: 1 whole, 1 whole pie, 1 pie, one)	When the numerator and denominator match, the fraction is 1 whole. 8/8 = 1 and 4/4 = 1, so they are the same amount.

Pre & Post sheet answer guide

Many items are open-ended. Grade for genuine understanding and complete sentences, not exact wording. Sheet: grade3/prepost/KCE_Math_PrePost_G3W28.pdf.

PRE-LESSON WARM-UP

#	Prompt	Model answer / grading note
1	Review: 8 by 6 patio with a 2 by 2 corner cut out	44 square feet. 8 × 6 = 48 and 2 × 2 = 4, and 48 − 4 = 44.

#	Prompt	Model answer / grading note
2	Draw 2 pies cut into 4 equal slices each; shade 6 slices	6 fourths (6/4). Check that each pie has 4 equal slices and 6 slices are shaded.
3	Circle the fraction equal to 1 whole pie (3/4, 4/4, 4/3)	4/4
4	Thirds in 1 whole pie; thirds in 2 whole pies	3 thirds make 1 whole pie; 6 thirds make 2 whole pies.
5	Is 5/4 of a pie more or less than 1 whole pie?	More. 4/4 is 1 whole pie, and 5/4 is 1 more fourth. Accept any reasonable explanation before the lesson.

POST-LESSON CHALLENGE

#	Prompt	Model answer / grading note
1	3 pies cut into sixths; write 3 with a denominator of 6	18 sixths, so 3 = 18/6 (3 × 6 = 18).
2	Number line from 0 to 3 in thirds; mark 4/3 and 8/3	The line shows 0, 1, 2 and 3 with 3 equal jumps in each whole. 4/3 is 1 jump past 1; 8/3 is 2 jumps past 2 (1 jump before 3).
3	Compare 7/6 and 7/8 with <, > or =	7/6 > 7/8. The numerators are the same, and sixths are bigger pieces than eighths. (7/6 is more than 1; 7/8 is less than 1.)
4	Declan says 15/3 of a pie is 15 whole pies	The 15 counts thirds, not whole pies. Count by 3s: 3, 6, 9, 12, 15. That is 5 whole pies, so 15/3 = 5.

★ CHALLENGE

#	Prompt	Model answer / grading note
1	13/4 pies of cherry pie: whole pies, leftover fourths, more or less than 3?	3 whole pies and 1/4 of a pie left over. 13/4 is more than 3, because 3 = 12/4 and 13 fourths is 1 more fourth.